

# Figure 4. Primordial tilt against 2025–26 CMB data

All estimates are base- $\Lambda$ CDM. Circles show the central estimate; bars show  $\pm 1\sigma$ .

— solid: dimension-zero  $n_s = 0.957888$

- - - dashed: amplitude branch  $n_s = 0.9498$

$n_s \pm 1\sigma$

vs solid ( $\sigma$ )

Planck 2018 TT,TE,EE+lowE+lensing

ACT DR6 alone

SPT-3G D1 alone

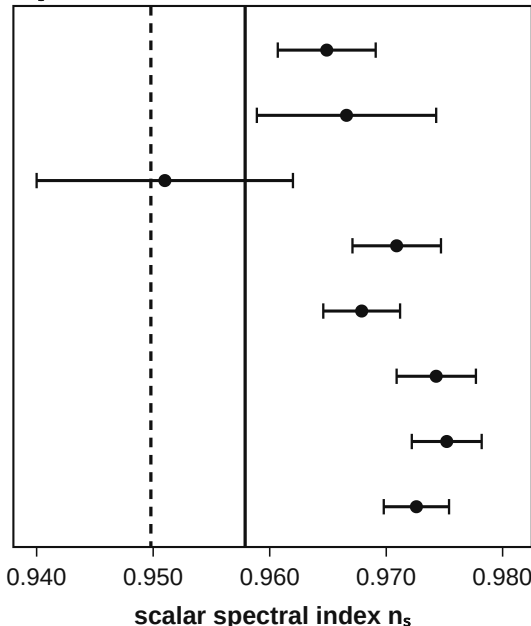
P-ACT

CMB-SPA

P-ACT-LB

P-ACT-LB2

CMB-SPA + DESI DR2



Prediction:  $n_s = 1 - 7\alpha_3(M_p)/\pi = 0.957888$ . Amplitude at  $N_0 = 36$  with Standard-Model couplings demands  $n_s = 0.9498$ .

The two values are not on the same one-parameter locus. This is posterior tension with the mechanism, not a formal exclusion.

The CPT-symmetric structure fixes no tilt. Freeing  $N_{\text{eff}}$  removes the tension, but needs  $\Delta N_{\text{eff}} \approx -0.35$  to  $-0.82$ .